1 Short Questions

1. $\boxed{12}$
(1) Applying L'Hospital's rule twice gives

$$\lim_{x\to 0}\frac{f(\sin^2 x)}{x^4}=\lim_{x\to 0}\frac{\sin 2x f'(\sin^2 x)}{4x^3}=\lim_{x\to 0}\frac{\sin^2(2x)f''(\sin^2 x)+2\cos 2x f'(\sin^2 x)}{12x^2}=\frac{12}{0}$$

which does not exist. Alternatively, use Taylor's expansion.
(2) Using the Taylor expansion $\sqrt{1+u}\approx 1+\frac{1}{2}u$,

$$\pi\sqrt{n^2+n}=\pi n\sqrt{1+\frac{1}{n}}=\pi n\left(1+\frac{1}{2n}-\frac{1}{8n^2}+\dots\right)=n\pi+\frac{\pi}{2}-\frac{\pi}{8n}+\mathcal{O}\left(\frac{1}{n^2}\right)$$

Substitute this into the sine function:

$$\sin^2\left(n\pi+\frac{\pi}{2}-\frac{\pi}{8n}+\dots\right)=\sin^2\left(\frac{\pi}{2}-\frac{\pi}{8n}+\dots\right)$$

As $n\to\infty$, the term inside goes to $\frac{\pi}{2}$. Hence,

$$\lim_{n\to\infty}\sin^2\left(\frac{\pi}{2}-\frac{\pi}{8n}+\dots\right)=\sin^2\left(\frac{\pi}{2}\right)=1$$

which exists.
(4) Given $F(x+\frac{z}{y},y+\frac{z}{x})=0$, we use implicit differentiation. Let $u=x+\frac{z}{y}$ and $v=y+\frac{z}{x}$. Differentiating $F(u,v)=0$ with respect to x and y: $F_u(1+\frac{1}{y}z_x)+F_v(-\frac{z}{x^2}+\frac{1}{x}z_x)=0\implies\frac{F_u}{F_v}=\frac{y(z-xz_x)}{x^2(y+z_x)}F_u(-\frac{z}{y^2}+\frac{1}{y}z_y)+F_v(1+\frac{1}{x}z_y)=0\implies\frac{F_u}{F_v}=\frac{y^2(x+z_y)}{x(z-yz_y)}$. Equating the two expressions for $\frac{F_u}{F_v}$:

$$\frac{y(z-xz_x)}{x^2(y+z_x)}=\frac{y^2(x+z_y)}{x(z-yz_y)}\implies(z-xz_x)(z-yz_y)=xy(y+z_x)(x+z_y)$$

Expanding both sides:

$$z^2-z(yz_y+xz_x)+xyz_xz_y=x^2y^2+xy(xz_x+yz_y)+xyz_xz_y$$

$$z^2-x^2y^2=(z+xy)(xz_x+yz_y)$$

$$(z-xy)(z+xy)=(z+xy)(xz_x+yz_y)$$

Dividing by $(z+xy)$, we get $xz_x+yz_y=z-xy$.
(8) Differentiatiate once:

$$3e^{3y}y'+(1+y')\cos[(x+y)^2]=0\implies y'=\frac{-\cos[(x+y)^2]}{3e^{3y}+\cos[(x+y)^2]}$$

Note that at $(0,0)$, $y'=-\frac{1}{4}$
Further differentiate:

$$3e^{3y}y''+9e^{3y}(y')^2-2(x+y)y'(1+y')\sin[(x+y)^2]+y''\cos[(x+y)^2]=0$$

Plugging in the numbers yields

$$3y'' + \frac{9}{16} + y'' = 0 \implies y'' = -\frac{9}{64}$$

2. $\boxed{3}$
The illustration using 2×2 matrix:
Let

$$A = \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix}$$

then

$$A^T = \begin{pmatrix} a_1 & a_3 \\ a_2 & a_4 \end{pmatrix}$$

The characteristic polynomial is

$$(a_1 - \lambda)(a_4 - \lambda) - a_2 a_3$$

for both A^T and A. They share same eigenvalues.
Consider the case that $A = \begin{pmatrix} 3 & 3 \\ 5 & 1 \end{pmatrix}$, then $A^T = \begin{pmatrix} 3 & 5 \\ 3 & 1 \end{pmatrix}$ We have

$$(3 - \lambda)(1 - \lambda) - 15 = 0 \implies \lambda = -2, 6$$

The eigenvectors for A are

$$\left\{ \begin{pmatrix} 3 \\ -5 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$$

The eigenvectors for A^T is

$$\left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 5 \\ 3 \end{pmatrix} \right\}$$

A^T and A do not share same eigenvectors.

3. $\boxed{\sin x + C \cos x}$
Differentiate once:

$$-f(x)\sin x + f'(x)\cos x + 2f(x)\sin x = 1$$
$$\implies f'(x) + f(x)\tan x = \sec x$$

Integrating factor:

$$\mu(x) = \exp\left(\int \tan x \, dx\right) = \exp[-\ln(\cos x)] = \sec x$$

Therefore,

$$\sec x f(x) = \int \sec^2 x \, dx = \tan x + C \implies f(x) = \sin x + C \cos x$$

4. $\boxed{15.9666817}$

n	x_n	f_n
0	0	0
1	0.2	0.739731083
2	0.4	2.840334703
3	0.6	8.648944725
4	0.8	25.6377146
5	1	83.93336676

Therefore, we have

$$\frac{0.2}{2}\left(f_0 + 2f_1 + 2f_2 + 2f_3 + 2f_4 + f_5\right) \approx 15.9666817$$

2 Long Questions

2.

$$\begin{aligned}
\mathcal{L}\{y''(t)\} &= \int_0^\infty e^{-st}y''(t)\,dt \\
&= \left[e^{-st}y'(t)\right]_0^\infty + s\int_0^\infty e^{-st}y'(t)\,dt \\
&= -y'(0) + s\left[e^{-st}y(t)\right]_0^\infty + s^2\int_0^\infty e^{-st}y(t)\,dt \\
&= s^2Y(s) - sy(0) - y'(0)
\end{aligned}$$

Dividing the first 2 terms of $\mathcal{L}\{y''(t)\}$ by s gives

$$\mathcal{L}\{y'(t)\} = sY(s) - y(0)$$

$$\begin{aligned}
\mathcal{L}\left\{\sin t\right\} &= \int_0^\infty e^{-st}\sin t\,dt \\
&= [-e^{-st}\cos t]_0^\infty - s\int_0^\infty e^{-st}\cos t\,dt \\
&= 1 - s[e^{-st}\sin t]_0^\infty - s^2\int_0^\infty e^{-st}\sin t\,dt \\
&= 1 - s^2\mathcal{L}\{\sin t\}
\end{aligned}$$

So

$$\mathcal{L}\{\sin t\} = \frac{1}{1+s^2}$$

The L.H.S. of the ODE becomes

$$s^2Y(s) - sy(0) - y'(0) + 3sY(s) - 3y(0) + 2Y(s) = (s^2+3s+2)Y(s) - y'(0) - sy(0) - 3y(0)$$

Pluging in the initial conditions yields

$$(s^2+3s+2)Y(s) - s - 3 = \frac{1}{1+s^2} \implies Y(s) = \frac{s+3}{(s+2)(s+1)} + \frac{1}{(s+2)(s+1)(s^2+1)}$$

Applying partial fraction decomposition on first term:

$$\frac{s+3}{(s+2)(s+1)} = \frac{p_1}{s+2} + \frac{p_2}{s+1} \implies s+3 = p_1(s+1) + p_2(s+2)$$

Put $s=-1$ gives $p_2 = 2$, put $s=-2$ gives $p_1 = -1$
Applying partial fraction decomposition on second term:

$$\frac{1}{(s+2)(s+1)(s^2+1)} = \frac{p_3}{s+2} + \frac{p_4}{s+1} + \frac{p_5s+p_6}{s^2+1}$$

$$\implies 1 = p_3(s+1)(s^2+1) + p_4(s+2)(s^2+1) + (p_5s+p_6)(s+2)(s+1)$$

Put $s=-1$ gives $p_4 = \frac{1}{2}$, put $s=-2$ gives $p_3 = -\frac{1}{5}$, put $s=0$ gives $p_6 = \frac{1}{10}$, thus $p_5 = -\frac{3}{10}$

Therefore,

$$Y(s) = \frac{5/2}{s+1} - \frac{6/5}{s+2} - \frac{3}{10}\frac{s}{s^2+1} + \frac{1/10}{s^2+1}$$

Applying inverse Laplace transform gives

$$y = \frac{5}{2}e^{-t} - \frac{6}{5}e^{-2t} - \frac{3}{10}\cos t + \frac{1}{10}\sin t$$

The general solution of the homogeneous equation is

$$y = C_1 e^{-x} + C_2 e^{-2x}$$

where C_1, C_2 are constants.

$$y_h' = A\cos t - B\sin t$$
$$y_h'' = -A\sin t - B\cos t$$

Matching gives

$$\begin{cases} A - 3B = 1 \\ 3A + B = 0 \end{cases} \implies A = \frac{1}{10}, B = -\frac{3}{10}$$

Applying initial conditions gives

$$\begin{cases} C_1 + C_2 - \frac{3}{10} = 1 \\ -C_1 - 2C_2 + \frac{1}{10} = 0 \end{cases} \implies C_1 = \frac{5}{2}, C_2 = -\frac{6}{5}$$

Which matches the result of Laplace transform.

3.

$$u \otimes (\alpha x + \beta y) = u\alpha x^T + u\beta y^T = \alpha u x^T + \beta u y^T = \alpha(u \otimes x) + \beta(u \otimes y)$$
$$(\alpha u + \beta v) \otimes x = (\alpha u + \beta v)x^T = \alpha u x^T + \beta v x^T = \alpha(u \otimes x) + \beta(v \otimes x)$$

Let

$$u = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}, x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

then

$$u \otimes x = \begin{pmatrix} u_1x_1 & u_1x_2 & u_1x_3 \\ u_2x_1 & u_2x_2 & u_2x_3 \\ u_3x_1 & u_3x_2 & u_3x_3 \end{pmatrix}$$

If this matrix is O, we need $u = 0\ \forall x \in M$. Similarly, we need $u = v$ if $u \otimes x = v \otimes x$.
Let

$$A = \begin{pmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{pmatrix}$$

Then applying

$$Ae_i = (u \otimes e_i)\mathbf{e}_i$$

we have

$$\begin{pmatrix} a_1 \\ a_4 \\ a_7 \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}, \begin{pmatrix} a_2 \\ a_5 \\ a_8 \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}, \begin{pmatrix} a_3 \\ a_6 \\ a_9 \end{pmatrix} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$$

which means all columns of A are equal to u.
We now change the notation:

$$A = \begin{pmatrix} p & p & p \\ q & q & q \\ r & r & r \end{pmatrix}$$

Then, we have

$$Ax = \begin{pmatrix} px_1 & px_2 & px_3 \\ qx_1 & qx_2 & qx_3 \\ rx_1 & rx_2 & rx_3 \end{pmatrix}$$

4. $\mathbf{g}$ preserves matrix addition, for example,

$$\begin{pmatrix} a & b \\ 0 & c \end{pmatrix} + \begin{pmatrix} a' & b' \\ 0 & c' \end{pmatrix} = \begin{pmatrix} a+a' & b+b' \\ 0 & c+c' \end{pmatrix}$$

It also preserves scalar multiplication, which is obvious.
We check for the conditions:

$$[\alpha x + \beta y, z] = \alpha xz + \beta yz - \alpha zx - \beta zy = \alpha[x,z] + \beta[y,z]$$
$$[z, \alpha x + \beta y] = \alpha zx + \beta zy - \alpha xz - \beta yz = \alpha[z,x] + \beta[z,y]$$
$$[x,y] = xy - yx = -(yx - xy) = -[y,x]$$

$$\begin{aligned} [x,[y,z]] + [y,[z,x]] + [z,[x,y]] &= x(yz - zy) - (yz - zy)x + y(zx - xz) - (zx - xz)y \\ &+ z(xy - yx) - (xy - yx)z \\ &= xyz - xzy - yzx + zyx + yzx - yxz \\ &- zxy + xzy + zxy - zyx - xyz + yxz \\ &= 0 \end{aligned}$$

Therefore, $(\mathbf{g}, [\cdot,\cdot])$ is a Lie algebra.

5. To derive the rotation matrix R_θ that rotates a vector about the origin by an angle θ, we consider the transformation of the standard basis $\mathcal{B} = \{\mathbf{e}_1, \mathbf{e}_2\}$, where $\mathbf{e}_1 = (1,0)^T$ and $\mathbf{e}_2 = (0,1)^T$.
Transform the basis vectors: Let T be a linear transformation that rotates space by θ counterclockwise. The image of $\mathbf{e}_1$ under rotation is $\mathbf{v}_1 = T(\mathbf{e}_1) = (\cos\theta, \sin\theta)^T$. The image of $\mathbf{e}_2$ is a vector perpendicular to $\mathbf{v}_1$ (rotated 90° further), which is $\mathbf{v}_2 = T(\mathbf{e}_2) = (-\sin\theta, \cos\theta)^T$.
Construct the Matrix: In linear algebra, the matrix representing a transformation relative to the standard basis is constructed by using the transformed basis vectors as columns.

$$R_\theta = [T(\mathbf{e}_1) \mid T(\mathbf{e}_2)] = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

The volume of revolution is

$$\pi \int_a^b [f(x)]^2 \, dx$$

Which can be derived by using cylindrical coordinates:
After revolution, we get a new function $r = f(z)$, then, we have

$$V = \int_0^{2\pi} \int_a^b \int_0^{f(z)} r \, drdzd\theta = \pi \int_a^b [f(z)]^2 \, dz = \pi \int_a^b [f(x)]^2 \, dx$$

We rotate the coordinate system by $\frac{\pi}{4}$, which gives

$$x' = \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}, y' = -\frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}$$

Then, in new coordinate system, $y \geq |x|$ is the first quadrant, $x + y \leq \frac{\pi\sqrt{2}}{4}$ is $x' \leq \frac{\pi}{4}$
Note that

$$x = \frac{1}{\sqrt{2}}(x' - y')$$

$$y = \frac{1}{\sqrt{2}}(x' + y')$$

For curve **S**, it becomes

$$\left(\frac{(x'-y')^2}{4} + \frac{(x'+y')^2}{4} - \frac{1}{2}(x'^2 - y'^2) \right)^2 = \tan x'$$

$$\implies y'^4 = \tan x'$$

The volume is

$$V = \pi \int_0^{\pi/4} \sqrt{\tan x} \, dx$$

Let $k = \sqrt{\tan u}$. Then $u = \arctan(k^2)$ and $du = \frac{2k}{1+k^4} dk$.At $u = 0, k = 0$; at $u = \pi/4, k = 1$.

$$V = \pi \int_0^1 \frac{2k^2}{1 + k^4} dk$$

By partial fraction,

$$\int_0^1 \frac{k^2+1}{k^4+1} dk = \frac{\pi}{2\sqrt{2}}, \quad \int_0^1 \frac{k^2-1}{k^4+1} dk = -\frac{1}{\sqrt{2}} \ln(\sqrt{2}+1)$$

Combining these results:

$$V = \pi \left[\frac{\pi}{2\sqrt{2}} - \frac{\ln(1+\sqrt{2})}{\sqrt{2}} \right] = \frac{\pi}{\sqrt{2}} \left(\frac{\pi}{2} - \ln(1+\sqrt{2}) \right)$$

6. **(b)** To show that if $y_1(t), y_2(t)$ are linearly independent, then $y_1(t)y_2'(t) \neq y_2(t)y_1'(t)$ for some t: Suppose for contradiction that $y_1y_2' - y_2y_1' = 0$ for all t. For $y_1(t) \neq 0$:

$$\frac{d}{dt}\left(\frac{y_2(t)}{y_1(t)}\right) = \frac{y_1y_2' - y_2y_1'}{y_1^2} = 0$$

Integrating both sides gives $\frac{y_2(t)}{y_1(t)} = k$, or $y_2(t) = ky_1(t)$. This implies y_1, y_2 are linearly dependent, contradicting the hypothesis.
To show that if $y_1(t)y_2'(t) \neq y_2(t)y_1'(t)$ for some t, then they are linearly independent: Consider $c_1y_1(t) + c_2y_2(t) = 0$. Differentiating gives $c_1y_1'(t) + c_2y_2'(t) = 0$. In matrix form:

$$\begin{pmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Since the determinant $W(t) = y_1y_2' - y_2y_1' \neq 0$ for some t, the matrix is invertible. The only solution is the trivial solution $c_1 = c_2 = 0$, proving linear independence.
(c) Define the Wronskian as $W(t) = y_1(t)y_2'(t) - y_2(t)y_1'(t)$. Differentiating with respect to t:

$$W' = y_1'y_2' + y_1y_2'' - y_2'y_1' - y_2y_1'' = y_1y_2'' - y_2y_1''$$

Since y_1 and y_2 satisfy $y'' + p(t)y' + q(t)y = 0$, substitute $y_i'' = -p(t)y_i' - q(t)y_i$:

$$W' = y_1(-py_2' - qy_2) - y_2(-py_1' - qy_1)$$

$$W' = -p(y_1y_2' - y_2y_1') = -p(t)W(t)$$

This is a first-order separable differential equation:

$$\frac{dW}{W} = -p(t)dt \implies \ln|W| = -\int p(t)dt + C$$

Exponentiating both sides:

$$y_1(t)y_2'(t) - y_2(t)y_1'(t) = A\exp\left(-\int p(t)dt\right)$$

(d) The system of equations for $u_1'(t)$ and $u_2'(t)$ is derived as

$$\begin{cases} u_1'(t)y_1(t) + u_2'(t)y_2(t) = 0 \\ u_1'(t)y_1'(t) + u_2'(t)y_2'(t) = g(t) \end{cases}$$

Solving this system using Cramer's Rule, where $W(y_1, y_2)(t)$ is the Wronskian determinant:

$$u_1'(t) = -\frac{y_2(t)g(t)}{W(y_1, y_2)(t)}$$
$$u_2'(t) = \frac{y_1(t)g(t)}{W(y_1, y_2)(t)}$$

Integrating these functions and substituting them into the trial solution $Y(t) = u_1(t)y_1(t) +$

$u_2(t)y_2(t)$ yields the general form:

$$Y(t) = -y_1(t)\int \frac{y_2(t)g(t)}{W(y_1,y_2)(t)}\,dt + y_2(t)\int \frac{y_1(t)g(t)}{W(y_1,y_2)(t)}\,dt$$

(e) is straightforward. For **(f)**, for the equation $t^2y'' + ty' + (t^2 - \frac{1}{4})y = 3t^{3/2}\sin t$: The homogeneous solutions are $y_1 = \frac{\sin t}{\sqrt{t}}$ and $y_2 = \frac{\cos t}{\sqrt{t}}$. The Wronskian is $W = -1/t$. The particular solution via variation of parameters is $y_p = -\frac{3\sqrt{t}\cos t}{2} + \frac{3\sin t}{4\sqrt{t}}$. The general solution is:

$$y(t) = C_1\frac{\sin t}{\sqrt{t}} + C_2\frac{\cos t}{\sqrt{t}} - \frac{3\sqrt{t}\cos t}{2}$$